

Fine tuning techniques

The bottleneck of giant models

- The Problem: Large Language Models (LLMs) have billions of parameters (\$7B\$, \$70B\$, etc.).
- The Cost: Full fine-tuning requires updating every parameter, demanding massive VRAM and compute.
- The VRAM Math: To fine-tune a \$7B\$ model in 16-bit, you need roughly 120GB–160GB of VRAM (far beyond a consumer GPU).
- The Goal: Find a way to teach a model new tricks without moving the "heavy furniture" of the original weights.

Problems with Full Fine-Tuning

- ☐ Requires **huge GPU memory**
- ☐ Slow
- ☐ Risk of catastrophic forgetting
- ☐ Not scalable for multiple tasks

LoRA Solves:

- ✓ Train **<1% of parameters**
- ✓ Much lower GPU memory
- ✓ Faster training
- ✓ Multiple LoRA adapters per base model

LoRA (Low-Rank Adaptation)

- **LoRA** is a **parameter-efficient fine-tuning (PEFT)** technique used to adapt large language models **without updating all model weights**.
- Instead of fine-tuning billions of parameters, LoRA:
 - **Freezes the base model**
 - **Injects small trainable low-rank matrices** into selected layers (usually attention layers)
- A large weight matrix **W** is expensive to update.
- LoRA assumes: “The *change* needed to adapt **W** is low-rank.”
- So instead of updating **W**, we learn:
 - $W' = W + \Delta W$
 - $\Delta W = A \times B$
- Where:
 - $A \in \mathbb{R}(d \times r)$
 - $B \in \mathbb{R}(r \times k)$
 - r is rank (very small: 4, 8, 16)
 - Only **A** and **B** are trained.

LoRA vs Full Fine-Tuning

Feature	Full FT	LoRA
Params trained	100%	<1%
Memory	Very high	Low
Speed	Slow	Fast
Overfitting	High risk	Lower
Multi-task	Hard	Easy

QLoRA (Quantized LoRA)

- QLoRA is an extension of LoRA that adds 4-bit quantization to the base model.
- Key idea: Quantize the frozen base model + apply LoRA adapters on top
- QLoRA Architecture:
 - Load base LLM in 4-bit
 - Freeze all base parameters
 - Inject LoRA adapters
 - Train only LoRA weights
 - Backprop happens through quantized weights safely
- **Why QLoRA Was Needed?**
 - Even LoRA **still requires loading the full base model in 16-bit.**
 - QLoRA enables:
 - Fine-tuning **65B models on a single 24GB GPU**
 - Democratization of LLM fine-tuning

LoRA vs QLoRA

Aspect	LoRA	QLoRA
Base model precision	FP16 / BF16	4-bit
Trainable params	Low-rank adapters	Same
GPU requirement	Medium	Very low
Performance	High	Near-LoRA
Complexity	Simple	Higher
Use case	Production fine-tuning	Resource-constrained setups